# Shayan Poigai

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## Experience

**Incoming SWE Intern @ HerbsPro**

**Co-Founder & Principal Software Engineer | Adorus (Jewelry E-commerce startup)** | *May 2026 - Present*

- Architecting and building a full-stack jewelry e-commerce platform from the ground up, leading both frontend and backend development
- Developing responsive frontend interfaces using React.js and Node.js
- Building RESTful backend services with Java Spring Boot and a PostgreSQL database
- Deploying and managing the application on AWS EC2 infrastructure

**Machine Learning Research Assistant | Santa Clara University** | *December 2025 — June 2026*

- Analyzing public sentiment toward using LLMs as a therapeutic tool, contributing to a multi-part academic paper
- Scraped and processed **5,000+ social media posts** from Twitter, Reddit, and YouTube across mental health and AI-related communities
- **Architected an NLP pipeline** to analyze 5000+ social media posts from Twitter, Reddit, and Youtube using **unsupervised topic modeling** (BERTopic, LDA, NMF)
- Optimized data preprocessing using **Pandas** to normalize unstructured text, reducing noise in high-dimensional social media datasets
- Created visual analytics (**Matplotlib, Seaborn**) revealing linguistic patterns and shifts in public attitudes toward AI-based mental health support
- Paper submitted to **ASONAM by IEEE 2026,** with results expected in mid-July 2026

**Computer Science Lab Teaching Assistant | Santa Clara University** | *September 2025 — March 2026*

- **Mentored 40+ students** in C++ memory management, pointers, and OOP principles, resulting in improved lab completion rates
- **Debug complex logic errors** in real-time, teaching students to utilize GDB and systematic testing to reduce technical debt in their coursework
- **Collaborated with faculty** to streamline lab delivery and ensure 100% alignment with core curriculum objectives

**Theory of Algorithms Grader | Santa Clara University |** *March 2026 — June 2026*

- **Graded assignments for 40+ students** across 2 sections of Theory of Algorithms

- Evaluated correctness and efficiency of solutions in **runtime analysis, divide & conquer, greedy algorithms, and dynamic programming**
- Provided clear, structured feedback and applied **consistent grading rubrics** to ensure fairness and improve student understanding

# Projects

**Disease Tracker | React/Node, FastAPI, AWS |** *October 2025*

- Developed a **FastAPI** RESTful API to serve real-time disease metrics and risk assessments with low-latency response times
- Integrated **AWS Bedrock** to engineer an automated, AI-assisted risk-scoring pipeline for public health data
- Designed **DynamoDB** NoSQL schemas to ensure scalable data ingestion and seamless backend-to-front-end communication

**Social Network Project | C++, QT |** *September 2025*

- Built a desktop social networking program in C++/QT with a GUI displaying user profiles, posts, and comments, applying MVC to separate UI from backend logic
- Implemented file I/O for persistent storage of users, friendships, posts, and comments, using OOP principles
- Modeled friendships with custom graph data structure and used BFS traversal to generate friend suggestions
- Automated multi-file builds with a Makefile

**Kuhn Poker vs Quantum** (Github | Live) **| Qiskit, IBM Quantum, FastAPI, React, Vercel, Render** | *June 2026*

- Built a poker game where users play against an opponent whose strategy is discovered by a 6-qubit quantum circuit using superposition and entanglement, optimized via Qiskit
- Developed a full-stack application with a FastAPI backend and React frontend, deployed on Render and Vercel
- Integrated the IBM Quantum API to execute the circuit on real quantum computing hardware and compare results against simulation

**Agentic RAG Pipeline | Python, ChromaDB, Ollama, MCP |** *July 2025*

- Engineered a modular **RAG framework** using **Ollama** and **ChromaDB** for semantic search across dynamic datasets
- Built an **MCP (Model Context Protocol) Server** to expose local data as tools for AI agents via **JSON-RPC**
- Designed a modular, extensible codebase that supports integration of new datasets with minimal refactoring

# Education & Skills

**Santa Clara University** | B.S. in Computer Science | *Expected Fall 2027* | **Major GPA: 4.0/4.0**

**Coursework**: Advanced Algorithms, Data Structures & Algorithms, Advanced C++, Linear Algebra, Theory of Automata, Discrete Mathematics

**Skills:** C++, Python, Java, SQL, NoSQL, FastAPI, SpringBoot, AWS, RAG, MCP, NLP, Docker, Linux, QT, React.js, Node.js, DynamoDB, ChromaDB, GitHub, OOP